

PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – Introduction to Programming

MODULE - 2

SESSION / ASSIGNMENT – 11

TASKS :

1. Declare an integer variable called likes and a pointer variable called ptrLikes; assign likes a value, point ptrLikes to likes, and print both the value and the address stored in ptrLikes.



```
#include <stdio.h>

int main()
{
    int likes = 500;
    int *ptrLikes = &likes;

    printf("\nValue of likes = %d\n", likes);
    printf("Value using pointer = %d\n", *ptrLikes);

    printf("\nAddress of likes = %p\n", &likes);
    printf("Address stored in ptrLikes = %p\n ", ptrLikes);

    return 0;
}
```

OUTPUT:

```
Value of likes = 500
Value using pointer = 500

Address of likes = 0061FF18
Address stored in ptrLikes = 0061FF18
```

2. Write a function `swapPlaylistCounts(int *a, int *b)` that swaps the number of songs in two Spotify playlists using pointers, then call the function in main and print the swapped values.



```
#include <stdio.h>

void swapPlaylistCounts(int *a, int *b)
{
    int temp;

    temp = *a;
    *a = *b;
    *b = temp;
}

int main()
{
    int playlist1 = 25;
    int playlist2 = 40;

    printf("Before Swap:\n");
    printf("Playlist 1 = %d\n", playlist1);
    printf("Playlist 2 = %d\n", playlist2);

    swapPlaylistCounts(&playlist1, &playlist2);

    printf("\nAfter Swap:\n");
    printf("Playlist 1 = %d\n", playlist1);
    printf("Playlist 2 = %d\n", playlist2);

    return 0;
}
```

OUTPUT:

```
Before Swap:
Playlist 1 = 25
Playlist 2 = 40

After Swap:
Playlist 1 = 40
Playlist 2 = 25
```

3. Given an array of 5 order amounts (e.g., Zomato orders), use a pointer to iterate through the array and print each amount along with its memory address
- Hint: Use pointer arithmetic to move to the next element.



```
#include <stdio.h>

int main()
{
    int orders[5] = {250, 400, 150, 300, 500};
    int *ptr = orders;
    int i;

    for(i = 0; i < 5; i++)
    {
        printf("Order Amount = %d\tAddress = %p\n", *ptr, ptr);
        ptr++;
    }

    return 0;
}
```

OUTPUT:

Order Amount = 250	Address = 0061FF04
Order Amount = 400	Address = 0061FF08
Order Amount = 150	Address = 0061FF0C
Order Amount = 300	Address = 0061FF10
Order Amount = 500	Address = 0061FF14

4. Create a function `incrementFollowers(int *followers, int n)` that increases each follower count in an array (representing Instagram followers for 5 friends) by 100 using pointer arithmetic, then print the updated counts.



```
#include <stdio.h>

void incrementFollowers(int *followers, int n)
{
    int i;

    for(i = 0; i < n; i++)
    {
        *(followers + i) += 100;
    }
}

int main()
{
    int followers[5] = {1200, 1500, 1800, 2000, 2500};
    int i;

    incrementFollowers(followers, 5);

    printf("Updated Followers:\n");

    for(i = 0; i < 5; i++)
    {
        printf("%d\n", followers[i]);
    }

    return 0;
}
```

OUTPUT:

```
Updated Followers:
1300
1600
1900
2100
2600
```

5. Draw a simple memory diagram (on paper or using any drawing tool) showing how a pointer references the address of a variable, using the variables from your likes and ptrLikes example; label the variable, pointer, and address.

